STAT 131: Lecture Notes 0

Winter 2024

0 Calculus Review

0.1 Review of Simple Series

Example 0.1. Evaluate $S = 1 + x + x^2 + x^3 + \cdots + x^n$

$(1-x)S = S - xS$

$= 1 + x + x^2 + x^3 + \cdots + x^n$

$- x - x^2 - x^3 - \cdots - x^{n+1}$

$= 1 - x^{n+1}$

$\Rightarrow S = \frac{1-x^{n+1}}{1-x}$ assuming $x \neq 1$

if $x = 1 \Rightarrow S = n+1$

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Example 0.2. Evaluate $S = 1 + x + x^2 + x^3 + \cdots$, where $|x| < 1$

$S = \lim_{n \to \infty} 1 + x + x^2 + x^3 + \cdots + x^n$

$= \lim_{n \to \infty} \frac{1-x^{n+1}}{1-x}$

$= \frac{1}{1-x}$

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Example 0.3. Evaluate $S = 1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1}$

$S = \frac{d}{dx}\left(1 + x + x^2 + x^3 + \cdots + x^n\right)$

$= \frac{d}{dx}\left(\frac{1-x^{n+1}}{1-x}\right) = \frac{(1-x)\left(-(n+1)x^n\right) + \left(1-x^{n+1}\right)}{(1-x)^2}$

$= \frac{1 - (n+1)x^n + nx^{n+1}}{(1-x)^2}$

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Example 0.4. Evaluate $S = 1 + 2x + 3x^2 + 4x^3 + \cdots$, where $|x| < 1$

$$S = \frac{d}{dx}\left(1 + x + x^2 + x^3 + \cdots\right)$$

$$= \frac{d}{dx}\left(\frac{1}{1-x}\right)$$

$$= \frac{1}{(1-x)^2}$$

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0.2 Integration

Definition 0.1. Integration by parts can be useful when we are integrating the product of two functions

$$\int_a^b u dv = (uv)\Big|_a^b - \int_a^b v du$$

Example 0.5 (Integration by Parts)**.** Find: $\int_0^\infty y e^{-y} dy$

Let $u = y$ and $dv = e^{-y} dy$

$du = dy$ $\quad$ $v = -e^{-y}$

$$\int_0^\infty y e^{-y} dy = -y e^{-y}\Big|_0^\infty - \int_0^\infty (-e^{-y}) dy$$

$$= (0 - 0) - \left(e^{-y}\Big|_0^\infty\right) = (0 - 0) - (0 - 1) = 1$$

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Definition 0.2 (Double Integrals)**.** Sometimes we may work with a double integral, where we are integrating with respect to two different variables. One approach is to just view this as a sequence of two integrations. We first integrate with respect to one variable, and then integrate again with respect to the other variable.

Example 0.6 (Double Integral)**.** Find: $\int_0^\infty \int_0^y e^{-y} dx dy$

$$\int_{y=0}^{y=\infty} \int_{x=0}^{x=y} e^{-y}\, dx\, dy = \int_{y=0}^{y=\infty} \left(x e^{-y} \Big|_{x=0}^{x=y} \right) dy$$

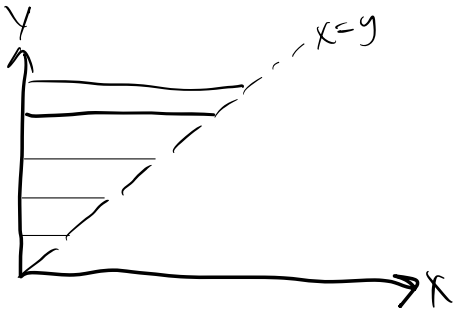
$$= \int_{y=0}^{y=\infty} y e^{-y}\, dy$$

$$= 1$$

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Definition 0.3 (Order of Integration)**.** When working with double integrals, we can also change the order of integration. If we do so, we need to first understand the space that we are integrating over.

Example 0.7 (Changing the Order)**.** Find: $\int_0^\infty \int_0^y e^{-y} dx dy$





$$\int_0^\infty \int_0^y e^{-y}\, dx\, dy = \int_0^\infty \int_x^\infty e^{-y}\, dy\, dx$$

$$= \int_0^\infty \left(-e^{-y} \Big|_{y=x}^{y=\infty} \right) dx$$

$$= \int_{x=0}^{x=\infty} \left(0 + e^{-x} \right) dx$$

$$= -e^{-x} \Big|_{x=0}^{x=\infty} = \boxed{1}$$

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Example 0.8. Find: $\int_1^e \int_0^{\ln(x)} 1 dy dx$

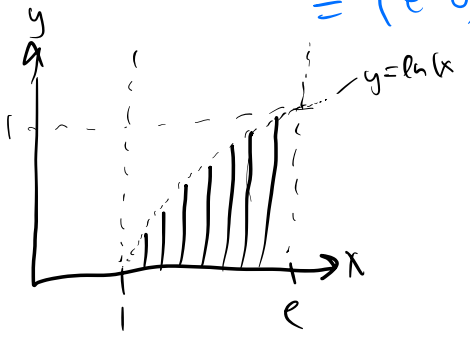
$u = \ln(x)$ $\quad dv = dx$

$du = \frac{1}{x} dx$ $\quad v = x$

$$\int_{x=1}^{x=e} \int_{y=0}^{y=\ln(x)} 1 \, dy \, dx = \int_{x=1}^{x=e} \ln(x) \, dx$$

$$= (\ln(x) \cdot x)\Big|_{x=1}^{x=e} - \int_{x=1}^{x=e} x \cdot \frac{1}{x} \, dx$$

$$= (e - 0) - (e - 1) = 1$$





$$\int_{y=0}^{y=1} \int_{x=e^y}^{x=e} 1 \, dx \, dy = \int_{y=0}^{y=1} \left(x \Big|_{x=e^y}^{x=e} \right) dy = \int_{y=0}^{y=1} (e - e^y) \, dy$$

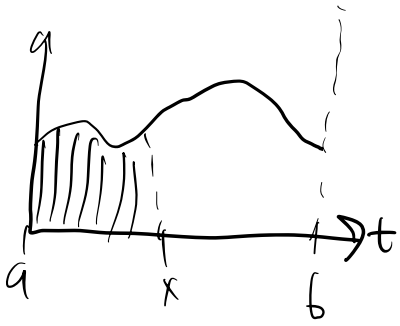
$$= (ey - e^y)\Big|_{y=0}^{y=1}$$

$$= (e - e) - (0 - 1) = 1$$

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Theorem 0.1 (Fundamental Theorem of Calculus)**.** *Let $f(t)$ be a continuous function defined on the interval $[a, b]$. Then for any x, where $a < x < b$,*

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$



$\int_a^x f(t) \, dt$ is the area under the curve from $t = a$ to $t = x$

We will use the fundamental theorem of calculus later in the course. We will not prove it, but instead focus on intuition here:

$$\text{Box}(x) = \int_a^x f(t)\,dt$$

$$\frac{d}{dx}\int_a^x f(t)\,dt = \frac{d}{dx}\text{Box}(x) = \lim_{\Delta\to 0}\frac{\text{Box}(x+\Delta) - \text{Box}(x)}{\Delta}$$

$$= \lim_{\Delta\to 0}\frac{\int_a^{x+\Delta} f(t)\,dt - \int_a^x f(t)\,dt}{\Delta}$$

$$= \lim_{\Delta\to 0}\frac{\int_x^{x+\Delta} f(t)\,dt}{\Delta} \sim \lim_{\Delta\to 0}\frac{f(x)\cdot\Delta}{\Delta} = f(x)$$

Theorem 0.2 (Extension of the FTC)**.** *For any differentiable function $g(x)$,*

$$\frac{d}{dx}\int_a^{g(x)} f(t)dt = f(g(x)))\cdot g'(x).$$

0.3 Review of Taylor Series and Limits

Definition 0.4 (Euler's number)**.** Euler's number or e is defined as a limit:

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e.$$

This comes up in many places and is a good thing to remember.

Example 0.9. If x is a constant, find

$$\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n.$$

Let $a=\frac{n}{x}$, then as $n\to\infty$ we also have $a\to\infty$

So $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n=\lim_{a\to\infty}\left(1+\frac{1}{a}\right)^{ax}=\lim_{a\to\infty}\left[\left(1+\frac{1}{a}\right)^a\right]^x=e^x$

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Definition 0.5 (Taylor Series)**.** For a function $f(x)$ that is infinitely differentiable at 0, the Taylor series expansion around 0 is

$$p(x)=f(0)+\frac{f'(0)}{1!}x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\cdots$$

Under the right conditions, $p(x)$ is a good approximation for $f(x)$ and and $p(x)$ will approach $f(x)$ as we use more and more terms. If we express $p(x)$ with an infinite number of terms, then $p(x)=f(x)$.

Example 0.10. Find an infinite expansion representation of the function $f(x) = e^x$.

$f(x) = e^x$, $f'(x) = e^x$, $f''(x) = e^x$, ...

$$e^x = f(0) + \frac{f'(0)\,x}{1!} + \frac{f''(0)\,x^2}{2!} + \frac{f'''(0)\,x^3}{3!} + \cdots$$

$$= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

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Example 0.11. Find an infinite expansion representation of the function $f(x) = e^{-x}$.

$f(x) = e^{-x}$, $f'(x) = -e^{-x}$, $f''(x) = e^{-x}$, ...

$$e^{-x} = f(0) + \frac{f'(0)\,x}{1!} + \frac{f''(0)\,x^2}{2!} + \frac{f'''(0)\,x^3}{3!} + \cdots$$

$$= 1 + \frac{-x}{1!} + \frac{x^2}{2!} + \frac{-x^3}{3!} + \cdots$$

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$$= 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \cdots$$